

TF-IDF Structured Ablation Results

TF-IDF Structured Ablation — Results & Parameters

Full results, parameters, and significance testing for the walk-forward comparison of **Structured** (baseline, no text) vs **Structured+TFIDF_full** (baseline + the fold's complete, uncapped TF-IDF vocabulary) vs **Structured+TFIDF_chi2** (baseline + top-500 TF-IDF columns selected by χ^2 , fit on train only), across Logistic Regression, XGBoost, and RandomForest. Companion to `analysis/BASELINE.py` (structured-only reference) and `FinBERT/RESULTS_COMPARISON.md` (the equivalent comparison for FinBERT embeddings).

1. Purpose

`TF-IDF/tfidf_pipeline.py` fits an uncapped (uncapped by feature-count cap; still `min_df=5` filtered) TF-IDF vocabulary per walk-forward fold (unigrams+bigrams) but does not train or evaluate any model on it. This script (`TF-IDF/tfidf_structured_ablation.py`) closes that gap: does adding TF-IDF improve default prediction over the structured-only baseline, and — since the whole point of using TF-IDF instead of a dense embedding is interpretability — how much of that benefit survives reducing the vocabulary to a small, nameable set of words/bigrams via χ^2 feature selection?

2. Data

- **Source CSV:** `data/03_advanced_prep/lc_after_03_advanced_prep_basic+test_20260715_2119.csv`
- **Rows after filtering** (`is_default ∈ {0,1}`): 217,287
- **Default rate:** 15.36%
- **Structured features:** 126 (identical `STRUCT_COLS` definition to `analysis/BASELINE.py`)
- **TF-IDF source:** `TF-IDF/output/fold{NN}/*.npz` — pre-computed by `tfidf_pipeline.py` (not recomputed here), `ngram_range=(1,2)`, `min_df=5`, `sublinear_tf=True`, uncapped vocabulary.

Walk-Forward folds (identical to `analysis/BASELINE.py` / `tfidf_pipeline.py`, asserted == 14)

`MIN_TRAIN_MONTHS=8`, `STEP_MONTHS=3`, `MIN_TEST_DEFAULTS=100` → 14 chronological expanding-window folds, 2010-08 to 2013-12.

Fold	train_cutoff	test_end	n_train	n_test	n_defaults
1	2010-08	2010-09	6,694	1,042	157
2	2010-11	2010-12	9,893	1,222	136
3	2011-02	2011-03	13,674	1,369	184
4	2011-05	2011-06	18,165	1,762	255
5	2011-08	2011-09	23,547	1,979	286
6	2011-11	2011-12	29,671	2,190	406
7	2012-02	2012-03	36,786	2,781	413
8	2012-05	2012-06	45,976	3,713	681
9	2012-08	2012-09	59,483	5,975	919
10	2012-11	2012-12	77,872	5,970	893
11	2013-02	2013-03	98,111	8,184	1,164
12	2013-05	2013-06	125,865	10,802	1,739
13	2013-08	2013-09	161,023	12,846	2,007
14	2013-11	2013-12	202,442	14,845	2,263

TF-IDF vocabulary size and runtime, per fold

The full arm's width grows with the fold (expanding-window training set → larger vocabulary passing `min_df=5`); the chi2 arm is fixed at 500 columns every fold. Runtime is the main practical cost of the full arm — RandomForest and XGBoost both fit on the complete sparse matrix (up to 55,028 TF-IDF columns + 126 structured, fold 14).

Fold	TF-IDF vocab (full arm)	Fold wall time (all 3 arms × 3 models)
1	6,227	55s
2	8,624	81s
3	11,226	101s
4	13,672	122s
5	16,504	156s
6	19,081	197s
7	21,482	248s
8	24,356	365s
9	29,073	524s
10	34,317	758s
11	39,208	1032s
12	44,235	997s
13	49,606	1362s
14	55,028	1788s

Total run time: 129.7 minutes (~2.2 hours), all 14 folds, run as a detached background process (`n_jobs=4` for RandomForest/XGBoost on the two TF-IDF arms, RAM safety on this 13.7GB machine).

3. Feature matrix construction (no leakage)

- **Structured block:** `SimpleImputer(strategy="constant", fill_value=0, keep_empty_features=True)`
 - `StandardScaler`, both fit on the fold's TRAIN rows only. (`keep_empty_features=True` was a bug fix during development — early folds have structured columns that are entirely NaN in that fold's small training slice; without it, `SimpleImputer` silently drops them, desyncing the feature-name list from the fitted model.)
- **Structured+TFIDF_full:** `scipy.sparse.hstack([structured_sparse, tfidf_train_full])` — the complete per-fold TF-IDF matrix, no reduction.
- **Structured+TFIDF_chi2:** `SelectKBest(chi2, k=500)` fit on the TF-IDF block of the TRAIN rows only (never touching the structured block — structured columns are never candidates for removal), then the identical selected columns applied to the TEST rows' TF-IDF matrix (train/test columns are index-aligned per fold since `tfidf_pipeline.py` fit the vectorizer on train then `.transform()` 'd test with the same vectorizer).
- All model fitting for the two TF-IDF arms stays **sparse** end-to-end (no densification) — `saga` (Logistic), `RandomForest`, and `XGBoost` all accept sparse CSR input natively.
- Feature names for TF-IDF-derived columns are prefixed `txt:` to avoid a name collision found during an earlier pipeline version (a structured column named e.g. `term` collided with a TF-IDF token also spelled `term`).

4. Parameters

Walk-forward / bootstrap

Param	Value
MIN_TRAIN_MONTHS	8
STEP_MONTHS	3
MIN_TEST_DEFAULTS	100
N_BOOTSTRAP	500
BOOT_SEED	42
RANDOM_STATE	242
CHI2_TOP_K	500

Fixed hyperparameters (identical across all 3 arms — reused verbatim from analysis/BASELINE.py , not re-tuned per arm)

Model	Params
Logistic	penalty=l2, C=0.3, solver=saga, max_iter=1000, class_weight=balanced
XGBoost	n_estimators=300, max_depth=4, learning_rate=0.05, subsample=0.8, scale_pos_weight=dynamic (neg/pos per fold), eval_metric=auc
RandomForest	n_estimators=500, min_samples_leaf=20, class_weight=balanced

Decided explicitly **not** to re-tune per arm (unlike FinBERT/finbert_structured_ablation.py , which tunes regularization per arm) — a full regularization sweep would multiply an already RAM/time-risky RandomForest fit on wide sparse data (up to ~55K columns) by 4–5 extra candidates × 3 tuning folds, for uncertain benefit relative to the added runtime (the actual run already took 130 minutes with fixed hyperparameters).

n_jobs : -1 for the Structured arm (small/dense, matches BASELINE.py exactly); 4 for the two TF-IDF arms (RAM safety on wide sparse RandomForest/XGBoost fits on this machine).

5. Results

5a. Full predictive performance — mean $\pm$ std across all 14 folds

Arm	Model	AUC	PR-AUC	Brier	GINI
Structured	Logistic	0.6930 $\pm$ 0.0152	0.2766 $\pm$ 0.0322	0.2313 $\pm$ 0.0151	0.3860 $\pm$ 0.0304
Structured	XGBoost	0.6904 $\pm$ 0.0191	0.2794 $\pm$ 0.0354	0.2114 $\pm$ 0.0171	0.3808 $\pm$ 0.0383
Structured	RandomForest	0.6882 $\pm$ 0.0199	0.2740 $\pm$ 0.0354	0.1891 $\pm$ 0.0142	0.3764 $\pm$ 0.0398
Structured+TFIDF_full	Logistic	0.6965 $\pm$ 0.0159	0.2818 $\pm$ 0.0318	0.2268 $\pm$ 0.0154	0.3930 $\pm$ 0.0318
Structured+TFIDF_full	XGBoost	0.6936 $\pm$ 0.0186	0.2827 $\pm$ 0.0273	0.2167 $\pm$ 0.0193	0.3872 $\pm$ 0.0373
Structured+TFIDF_full	RandomForest	0.6824 $\pm$ 0.0183	0.2700 $\pm$ 0.0289	0.2268 $\pm$ 0.0127	0.3647 $\pm$ 0.0366
Structured+TFIDF_chi2	Logistic	0.6942 $\pm$ 0.0164	0.2791 $\pm$ 0.0307	0.2308 $\pm$ 0.0151	0.3884 $\pm$ 0.0328
Structured+TFIDF_chi2	XGBoost	0.6936 $\pm$ 0.0204	0.2799 $\pm$ 0.0298	0.2166 $\pm$ 0.0173	0.3872 $\pm$ 0.0408
Structured+TFIDF_chi2	RandomForest	0.6870 $\pm$ 0.0210	0.2753 $\pm$ 0.0329	0.2069 $\pm$ 0.0137	0.3740 $\pm$ 0.0420

5b. DeLong pairwise significance test (paired, same test set, per fold)

Model	Comparison (B vs A)	Mean Δ AUC (B-A)	Significant folds ($p < 0.05$)
Logistic	Structured+TFIDF_full vs Structured	+0.0035	7/14
Logistic	Structured+TFIDF_chi2 vs Structured	+0.0012	2/14
Logistic	Structured+TFIDF_chi2 vs Structured+TFIDF_full	-0.0023	8/14
XGBoost	Structured+TFIDF_full vs Structured	+0.0032	3/14
XGBoost	Structured+TFIDF_chi2 vs Structured	+0.0032	2/14
XGBoost	Structured+TFIDF_chi2 vs Structured+TFIDF_full	+0.0000	0/14
RandomForest	Structured+TFIDF_full vs Structured	-0.0058	7/14
RandomForest	Structured+TFIDF_chi2 vs Structured	-0.0012	3/14
RandomForest	Structured+TFIDF_chi2 vs Structured+TFIDF_full	+0.0046	6/14

5c. Per-model findings

- **Logistic:** TF-IDF (full) improves on Structured ($\Delta\text{AUC}=+0.0035$, significant in 7/14 folds). Reducing to the top-500 chi^2 -selected words recovers most of the full arm's benefit (chi^2 vs Structured $\Delta\text{AUC}=+0.0012$, significant in 2/14 folds; full vs chi^2 $\Delta\text{AUC}=+0.0023$, significant in 8/14 folds). Best-performing arm overall: Structured+TFIDF_full.
 - **XGBoost:** TF-IDF (full) improves on Structured ($\Delta\text{AUC}=+0.0032$, significant in 3/14 folds). Reducing to the top-500 chi^2 -selected words recovers essentially all the full arm's benefit (chi^2 vs Structured $\Delta\text{AUC}=+0.0032$, significant in 2/14 folds; full vs chi^2 $\Delta\text{AUC}=-0.0000$, significant in 0/14 folds). Best-performing arm overall: Structured+TFIDF_chi2.
 - **RandomForest:** TF-IDF (full) does not improve on Structured ($\Delta\text{AUC}=-0.0058$, significant in 7/14 folds). Reducing to the top-500 chi^2 -selected words recovers most of the full arm's benefit (chi^2 vs Structured $\Delta\text{AUC}=-0.0012$, significant in 3/14 folds; full vs chi^2 $\Delta\text{AUC}=-0.0046$, significant in 6/14 folds). Best-performing arm overall: Structured.
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6. Which TF-IDF words actually contributed (chi2 arm, all 14 folds)

Cross-fold consistency: for each model, which words were selected by chi^2 and how highly the model itself (not the chi^2 statistic) ranked them, across all 14 independently-trained folds. A word appearing in most/all folds, with a consistently high importance rank, is the strongest candidate to carry into a non-walk-forward final model — it was re-derived independently each time, not chosen once on the full dataset.

Logistic — top 20 most consistent words

Word	Folds selected	Mean rank	Mean importance share	Mean chi2 score
help	14/14 (100%)	17.8	0.0060	4.59
ring	14/14 (100%)	18.2	0.0058	5.51
ups	14/14 (100%)	20.8	0.0055	5.74
engagement	14/14 (100%)	25.3	0.0052	6.45
engagement ring	14/14 (100%)	42.9	0.0042	5.18
apr	13/14 (93%)	9.5	0.0077	5.20
pool	13/14 (93%)	25.6	0.0092	11.85
mta	13/14 (93%)	65.4	0.0040	10.44
motorcycle	13/14 (93%)	108.6	0.0028	3.75
help foot	13/14 (93%)	293.8	0.0013	4.41
trailer	12/14 (86%)	53.9	0.0036	2.72
excellent credit	12/14 (86%)	104.6	0.0026	3.43
pay problem	12/14 (86%)	149.2	0.0020	3.19
buster	12/14 (86%)	160.8	0.0020	3.53
payday	12/14 (86%)	219.1	0.0018	3.31
college	11/14 (79%)	2.5	0.0119	4.57
university	11/14 (79%)	6.7	0.0105	6.33
walmart	11/14 (79%)	45.1	0.0043	5.33
harbor	11/14 (79%)	73.9	0.0031	4.01
good payer	11/14 (79%)	127.8	0.0022	7.58

XGBoost — top 20 most consistent words

Word	Folds selected	Mean rank	Mean importance share	Mean chi2 score
engagement	14/14 (100%)	12.7	0.0065	6.45
ring	14/14 (100%)	17.1	0.0062	5.51
help	14/14 (100%)	67.3	0.0045	4.59
ups	14/14 (100%)	106.9	0.0040	5.74
engagement ring	14/14 (100%)	220.8	0.0021	5.18
motorcycle	13/14 (93%)	23.2	0.0057	3.75
apr	13/14 (93%)	37.9	0.0053	5.20
pool	13/14 (93%)	73.0	0.0052	11.85
mta	13/14 (93%)	121.2	0.0040	10.44
help foot	13/14 (93%)	257.8	0.0011	4.41
excellent credit	12/14 (86%)	103.2	0.0040	3.43
pay problem	12/14 (86%)	106.4	0.0039	3.19
trailer	12/14 (86%)	107.1	0.0041	2.72
payday	12/14 (86%)	222.8	0.0022	3.31
buster	12/14 (86%)	440.2	0.0003	3.53
college	11/14 (79%)	15.4	0.0063	4.57
university	11/14 (79%)	17.7	0.0061	6.33
refinance	11/14 (79%)	21.8	0.0057	5.39
harbor	11/14 (79%)	107.3	0.0039	4.01
good payer	11/14 (79%)	148.7	0.0032	7.58

RandomForest — top 20 most consistent words

Word	Folds selected	Mean rank	Mean importance share	Mean chi2 score
help	14/14 (100%)	34.6	0.0073	4.59
ring	14/14 (100%)	79.6	0.0010	5.51
engagement	14/14 (100%)	80.4	0.0010	6.45
engagement ring	14/14 (100%)	93.1	0.0006	5.18
ups	14/14 (100%)	195.9	0.0000	5.74
apr	13/14 (93%)	55.2	0.0028	5.20
pool	13/14 (93%)	78.5	0.0013	11.85
motorcycle	13/14 (93%)	79.0	0.0010	3.75
mta	13/14 (93%)	263.6	0.0000	10.44
help foot	13/14 (93%)	278.5	0.0000	4.41
excellent credit	12/14 (86%)	70.3	0.0019	3.43
trailer	12/14 (86%)	192.1	0.0001	2.72
payday	12/14 (86%)	223.8	0.0000	3.31
pay problem	12/14 (86%)	252.2	0.0000	3.19
buster	12/14 (86%)	435.8	0.0000	3.53
college	11/14 (79%)	51.9	0.0030	4.57
refinance	11/14 (79%)	55.6	0.0022	5.39
university	11/14 (79%)	57.2	0.0022	6.33
walmart	11/14 (79%)	121.5	0.0002	5.33
harbor	11/14 (79%)	289.7	0.0000	4.01

Candidate features for a final model: the words above selected in 12+/14 folds across all three models (help , ring , engagement , engagement ring , ups , apr , pool , motorcycle , mta , excellent credit) are the most defensible TF-IDF features to carry forward into a non-walk-forward final model.

7. Output file locations

TF-IDF/results/ablation_folds.csv	per fold x arm x model: AUC/PR-AUC/Brier/GINI + 95% CI
TF-IDF/results/ablation_summary.csv	mean +/- std across folds, grouped by (arm, model)
TF-IDF/results/ablation_delong.csv	pairwise DeLong per fold x model x arm-pair
TF-IDF/results/ablation_feature_importance.csv	per fold x arm x model x feature: importance/rank
TF-IDF/results/ablation_feature_consistency.csv	cross-fold stability of the chi2 arm's selected words

TF-IDF/results/ablation_report.md	this document
TF-IDF/figures/fig1_auc_by_fold.png	AUC per fold, per model, 3 arms overlaid
TF-IDF/figures/fig2_dealong_deltas.png	Δ AUC bar chart per model x arm-pair, significance-colored